Parisa Gunasekhar

+91 8885522517 | gunasekharparisa2218@gmail.com | Linkedin - gunasekhar-parisa | Github - imguna2218 | Portfolio

Education

Lakireddy Bali reddy college of Engineering CGPA: 8.88

Bachelor of Technology in Computer Science and Engineering *Sep 2022 – Jul 2026*

Summary

Software Engineer specializing in distributed backend infrastructure (Python, Java, Rust). Proven experience driving architectural design for ambiguous technical problems, building fault-tolerant microservices, and optimizing algorithms to deliver highly scalable, high-concurrency enterprise systems.

Work Experience

Software Engineer | *Aksi Aerospace Group* Mar 2026 – Present

- Led the end-to-end architectural design of an automated data processing pipeline from broadly defined requirements, utilizing **Python (FastAPI)** and **SQL** to unify disparate subsystems and process large-scale datasets.
- Designed the **backend architecture** and **PostgreSQL** data schemas for a 4-tier billing system, developing scalable **RESTful APIs** secured with payload encryption and idempotency keys.
- Resolved access control ambiguities by designing a custom **OAuth 2.0** schema in **MongoDB**, successfully integrating **Multi-Factor Authentication (MFA)** across the core enterprise application.
- Authored comprehensive **API specifications and technical documents**, clarifying ambiguous business logic to align engineering teams on pipeline integrations and operational workflows.
- Debugged state desynchronization during peak workloads by designing a cross-service logging and **system telemetry** framework, stabilizing real-time event updates via a decoupled result queue.

Software Developer Intern | *LeanTech Labs Pvt. Ltd.* Jan 2025 – Sep 2025

- Applied fundamental **system design principles** to build a modular **vendor management** solution for an enterprise LMS, developing responsive, reusable frontend components using **React.js** and robust **REST APIs** in Node.js.
- Actively contributed to product quality improvements within an **Agile squad**, conducting peer code reviews and comprehensive unit testing via **Test-Driven Development (TDD)** to enforce secure coding standards.
- **Debugged and resolved critical application defects** across the stack, analyzing system behavior to prevent potential **service degradation and downtime** to ensure operational stability.

Skills

Languages: Java, Python, JavaScript (ES6+), TypeScript, Rust, C

Backend & Frontend: Spring Boot, FastAPI, NodeJS, ExpressJS, NestJS, React.js, NextJS, HTML5, CSS3, REST APIs

Databases & Tools: PostgreSQL, MongoDB, SQL, PL/SQL Kafka, Redis, Postman, VS Code, Git, GitHub, Data Structures

Cloud & AI: AWS, Docker, Kubernetes, CI/CD, Generative AI, LLMs, AI Agents, RAG.

Projects

EvalX – An Online Code Judge System | *Rust, Isolate, Docker, Amazon SQS*

- **Engineered a distributed, event-driven microservices code evaluation engine** in Rust on **Ubuntu**, optimizing concurrency algorithms and memory allocation to isolate and process **120+ concurrent test cases per second**.
- Architected a fault-tolerant storage and querying layer using **PostgreSQL**, integrating **Amazon SQS** and **Redis** to manage asynchronous workload queues and decouple execution state during peak traffic.
- Automated **AWS** deployments via **CI/CD** pipelines and authored API documentation for LMS integration.

IgnisOps – AI Powered Cloud Development Environment | *Node.js, Firecracker, NextJS, Websockets*

- Engineered a **NextJS/React** frontend for provisioning workspaces, and architected a secure **Node.js** backend using Firecracker MicroVMs to execute multi-language workloads with **sub-200ms cold starts**.
- Engineered **backend APIs** for a **3-tier AI pipeline**, designing a distributed data layer utilizing **PostgreSQL** for persistent storage and **Redis** for rapid session recovery and context-aware state management.

Certifications

AWS Cloud Practitioner *AWS*
Certified Application Developer *ServiceNow*
TECH A MERN STACK Certification *Infosys SpringBoard*

Achievements and Co-curricular Activities

- Won **1st Prize** at the Siddhartha Academy of Higher Education's Innovation Day competition for the **EvalX** project.
- Won **1st prize** in VR Siddhartha International Hackathon, competing in the **Code Geeks** track.
- Won **1st prize** in HackIT conducted in our college in the domain of **WEB DEVELOPMENT**.
- Solved 500+ algorithmic and data structure problems on LeetCode (Java), earning the 100-day badge twice.